

OPERATIONAL SUSTAINABILITY

Sustainable Supply Chains

Integrating environmental considerations into sourcing, production, and distribution

**You already know 70–90% of a company's footprint lives in its
supply chain.
Today is about what to actually do about it.**

*Three stages — sourcing, production, distribution — and the specific interventions, standards, and case studies that
make each one measurably more sustainable.*

Following the Product, Upstream to Downstream

Sustainable Supply Chain Management (SSCM): integrating environmental and social criteria into every decision along a product's journey — not just at the factory gate.



01

Sourcing

Where raw materials come from, and under what conditions.



02

Production

How materials become products — and at what environmental cost.



03

Distribution

How products move to market — often the most visible emissions of all.



04

Measuring the Chain

Mapping, tracking, and reporting the whole system end to end.

SECTION 1 OF 4

Sustainable Sourcing

Where raw materials come from, and under what conditions — the origin point of most supply chain risk.



What Is Sustainable Sourcing?

The furthest-upstream link in the chain — and, per the GHG Protocol, usually the largest share of a company's real footprint.

The Working Definition

Procuring raw materials and components in ways that account for environmental impact, labor conditions, and long-term resource availability — not price alone.

Why Upstream Matters Most

Categories 1 and 4 of Scope 3 (purchased goods, upstream transportation) are frequently a company's single largest emissions category — often larger than its own operations.

The Core Challenge

A company rarely owns its suppliers — sustainable sourcing means influencing behavior it cannot directly control, several tiers removed from its own operations.

SCOPE 3: Categories:

Everything that happens before a product or service reaches your organization.



1. Purchased Goods & Services

Emissions from producing everything your organization buys or acquires.



2. Capital Goods

Emissions from producing purchased long-term assets — machinery, buildings, equipment.



3. Fuel- & Energy-Related Activities

Upstream emissions of fuel and energy not already counted in Scope 1 or 2.



4. Upstream Transportation & Distribution

Third-party transport of purchased products to your organization.



5. Waste Generated in Operations

Emissions from third-party disposal and treatment of your operational waste.



6. Business Travel

Employee travel for business in vehicles not owned by the organization — flights, trains, taxis.



7. Employee Commuting

Emissions from employees' travel between home and work.



8. Upstream Leased Assets

Emissions from operating assets you lease, not already in Scope 1 or 2.

Four Categories of Sourcing Risk

The specific failure modes that make upstream sourcing the riskiest link in most supply chains.



Deforestation-Linked Commodities

Palm oil, soy, cattle, and timber are consistently linked to tropical deforestation when sourced without traceability.



Conflict Minerals

Tin, tungsten, tantalum, and gold sourced from conflict-affected regions can finance armed groups and human rights abuses.



Forced & Child Labour

Complex, multi-tier supply chains can obscure labor abuses far removed from the buying company's direct view.



Biodiversity & Habitat Loss

Extraction of raw materials — mining, fishing, logging — can degrade ecosystems far from where the final product is sold.

Certifications That Address Sourcing Risk

Third-party certification is the primary tool buyers use to verify claims they cannot check themselves.



Fairtrade

Certifies fair prices and labor standards for commodities like coffee, cocoa, and cotton, verified by independent audit.



Forest Stewardship Council (FSC)

Certifies wood and paper products from responsibly managed forests, with full chain-of-custody tracking.



Rainforest Alliance

Certifies farms meeting environmental, social, and economic sustainability criteria, common in coffee, tea, and cocoa.



Responsible Minerals Initiative

Audits smelters and refiners to verify minerals are not sourced from conflict-affected or high-risk areas.

When Sourcing Oversight Fails: Rana Plaza



The Rana Plaza Collapse, Dhaka, Bangladesh (April 24, 2013)

An eight-story building housing five garment factories collapsed, killing 1,134 workers and injuring approximately 2,500 more — the deadliest disaster in the history of the garment industry.

The Sourcing Failure

The factories supplied major global apparel brands, whose supplier audits had never identified the building's fundamental structural defects.

The Warning Ignored

Visible cracks were discovered the day before the collapse; workers were ordered back to their 13–14 hour shifts regardless.

The Systemic Response

The legally binding Accord on Fire and Building Safety in Bangladesh was signed within a month by over 200 global brands and retailers.

The Lesson

Supplier audits that check labor and quality standards, but not structural safety, can miss the most catastrophic risks entirely.

"A brand's supplier audit checks fire exits and pay stubs, but not structural engineering. What other kinds of 'invisible' risk might a standard audit checklist miss today?"

Think beyond garments — electronics, food, minerals.

📄 3 MIN

For most companies, which GHG Protocol Scope 3 categories are most directly addressed by sustainable sourcing practices?

A Category 11 (Use of sold products) and Category 12 (End-of-life)

B Category 1 (Purchased goods & services) and Category 4 (Upstream transportation) ✓

C Category 6 (Business travel) only

D Sourcing has no connection to Scope 3 at all

ANSWER & WHY

B. Category 1 (Purchased goods & services) and Category 4 (Upstream transportation)

Sustainable sourcing addresses the upstream categories of Scope 3 most directly — Category 1 (purchased goods & services) and Category 4 (upstream transportation) — which are frequently a company's single largest emissions categories.

SECTION 2 OF 4

Sustainable Production

How raw materials become products — and the environmental cost embedded in every design choice.



What Is Sustainable Production?

Manufacturing that minimizes environmental impact per unit of output — not simply "cleaner factories."

The Working Definition

Producing goods using processes and materials that minimize energy use, water use, waste, and toxic emissions relative to the value created.

Cleaner Production, Not End-of-Pipe

The modern approach prevents pollution and waste at the source — in process and material design — rather than treating waste after it's already created.

Why It's a Design Problem

The vast majority of a product's environmental impact is locked in at the design stage, long before a single unit is manufactured.

Life Cycle Assessment (LCA)

The standardized method for actually measuring a product's environmental impact — from raw material to disposal.

Cradle-to-Gate

Measures impact from raw material extraction through to the factory gate — the boundary most manufacturers can measure directly.

Cradle-to-Grave

Extends the assessment through use and disposal — the full picture of a product's lifetime impact.

Cradle-to-Cradle

Extends further still: designed explicitly so the product's materials become the raw material for the next product cycle.

Why It Matters Here

LCA is what lets a company compare two production or material choices quantitatively, instead of guessing which is "greener."

SECTION 2 · WORKED EXAMPLE

Comparing Packaging Materials with LCA Thinking

GIVEN

- Two packaging options, each weighing 50 g (0.050 kg) per unit
- Option A — Virgin plastic: embodied emissions of 3.5 kgCO₂e per kg of material produced
- Option B — Recycled plastic (rPET): embodied emissions of 1.2 kgCO₂e per kg of material produced

FORMULA

$$\text{Embodied Emissions per Unit} = \text{Weight (kg)} \times \text{Emission Factor (kgCO}_2\text{e/kg)}$$

STEP-BY-STEP CALCULATION

- 1 Option A: $0.050 \text{ kg} \times 3.5 \text{ kgCO}_2\text{e/kg} = 0.175 \text{ kgCO}_2\text{e per unit}$
- 2 Option B: $0.050 \text{ kg} \times 1.2 \text{ kgCO}_2\text{e/kg} = 0.060 \text{ kgCO}_2\text{e per unit}$
- 3 Savings per unit: $0.175 - 0.060 = 0.115 \text{ kgCO}_2\text{e}$
- 4 At 1 million units produced per year: $0.115 \text{ kg} \times 1,000,000 = 115,000 \text{ kg} = 115 \text{ tonnes CO}_2\text{e saved}$

115 t CO₂e / Year

A single material substitution — virgin to recycled plastic — saves the equivalent of many households' annual emissions, without changing the product itself.

Four Cleaner Production Techniques

Concrete interventions that reduce environmental impact at the point of manufacture.



Energy & Water Efficiency

Optimizing industrial processes to use less energy and water per unit of output — often the fastest payback in a factory.



Waste Minimization

Redesigning processes to generate less scrap, off-cuts, and by-product in the first place — not just recycling more of it.



Safer Chemical Substitution

Replacing hazardous solvents, dyes, and inputs with less toxic alternatives that achieve the same manufacturing result.



Process Optimization

Fine-tuning temperatures, cycle times, and equipment settings to cut energy and material use without changing the product.

What is the key difference between "cradle-to-gate" and "cradle-to-cradle" in Life Cycle Assessment?

- A They measure exactly the same boundary
- B Cradle-to-gate stops at the factory; cradle-to-cradle is designed so end-of-life materials become new raw material ✓**
- C Cradle-to-cradle only applies to food products
- D Cradle-to-gate includes disposal, cradle-to-cradle does not

ANSWER & WHY

B. Cradle-to-gate stops at the factory; cradle-to-cradle is designed so end-of-life materials become new raw material

Cradle-to-gate measures impact only up to the factory gate. Cradle-to-cradle is the most ambitious boundary — the product and its materials are explicitly designed to become the raw material for a future product cycle, embodying circular economy principles directly in the LCA scope.

SECTION 3 OF 4

Sustainable Distribution

How products move to market — often the most visible, and most easily reducible, link in the chain.



Distribution's Environmental Footprint

~1/5

Share of global CO₂ emissions attributable to the transport sector as a whole

4–6

Category 4 (upstream) and Category 9 (downstream) transportation — two of the most under-measured Scope 3 categories

75×

How much more emissions-intensive air freight is than sea freight, for the same cargo and distance

Distribution choices are unusual in sustainability: the lowest-carbon option is frequently also the lowest-cost option — a rare true win-win.

Source: Illustrative figures based on published freight-transport emission factors (e.g., DEFRA, ICCT).

Comparing Freight Transport Modes

GIVEN

- Shipment: 20 tonnes of goods, travelling 6,000 km
- Sea freight emission factor: ~10 gCO₂e per tonne-km
- Air freight emission factor: ~600 gCO₂e per tonne-km

FORMULA

$$\text{Emissions (kg)} = \text{Weight (t)} \times \text{Distance (km)} \times \text{Emission Factor} \div 1,000$$

STEP-BY-STEP CALCULATION

- 1 Sea freight: $20 \times 6,000 \times 10 \div 1,000 = 1,200 \text{ kg} \approx 1.2 \text{ tonnes CO}_2\text{e}$
- 2 Air freight: $20 \times 6,000 \times 600 \div 1,000 = 72,000 \text{ kg} = 72 \text{ tonnes CO}_2\text{e}$
- 3 Ratio: $72 \div 1.2 = 60\times$ more emissions for the identical shipment, moved by air instead of sea

60× Difference

The mode choice alone — sea versus air, for the exact same cargo — changes the shipment's footprint by a factor of 60. Mode shift is one of the highest-leverage decisions in distribution.

Five Levers of Sustainable Distribution

Interventions that reduce distribution's footprint without necessarily changing what is sold.



Modal Shift

Moving freight from air or road to sea or rail wherever transit time allows — the single highest-leverage lever available.



Route Optimization

Software-driven routing that reduces total distance travelled and empty return trips ("deadheading").



Packaging Reduction

Lighter, right-sized packaging reduces both material use and the weight (and therefore fuel) needed to ship it.



Warehouse Efficiency

Locating distribution centers closer to demand, and using energy-efficient warehouse operations, cuts both distance and energy use.



Reverse Logistics

Efficient systems for returns, repairs, and end-of-life collection keep products and materials circulating rather than being discarded.

Sourcing, Production, or Distribution?

For each real intervention, decide which stage of the supply chain it primarily belongs to.

1 Requiring FSC certification for all wood suppliers

2 Switching a factory from virgin to recycled plastic input

3 Shifting freight from air to sea for non-urgent shipments

4 Auditing a mineral supplier against conflict-minerals standards

5 Redesigning a manufacturing process to cut water use

6 Right-sizing packaging to reduce shipping weight

SOURCING**PRODUCTION****DISTRIBUTION**

1

2

3

4

5

6

In the worked example, why did the air freight shipment produce roughly 60 times more emissions than the identical sea freight shipment?

- A The air shipment travelled a much longer distance
- B The air shipment carried more weight
- C Air freight has a much higher emission factor (gCO₂e per tonne-km) than sea freight ✓**
- D There is no real difference between air and sea freight emissions

ANSWER & WHY

C. Air freight has a much higher emission factor (gCO₂e per tonne-km) than sea freight

Both shipments had identical weight and distance — the entire 60× difference comes from the emission factor itself: air freight emits far more CO₂e per tonne-kilometre than sea freight, due to the energy intensity of air transport.

SECTION 4 OF 4

Measuring & Managing the Chain

You cannot manage what you cannot see — the visibility problem underneath every claim in this lecture.

Supply Chain Tiers: Why Visibility Collapses Fast

Most companies can see their direct suppliers clearly — and almost nothing beyond that.

Tier 1

Direct suppliers the company contracts with and pays directly.

Usually well audited and visible.

Tier 2

Suppliers to your Tier 1 suppliers — e.g., the fabric mill supplying a garment factory.

Visibility drops sharply here.

Tier 3+

Raw material extraction — mines, farms, forests — often multiple contractual layers removed.

Frequently invisible to the buying company entirely.

The Whole Lecture Is a Scope 3 Map

Every stage covered today maps directly onto specific GHG Protocol Scope 3 categories.

Sourcing

Cat. 1 — Purchased goods & services · Cat. 2 — Capital goods

Production

Cat. 3 — Fuel- & energy-related activities · Cat. 5 — Waste generated in operations

Distribution

Cat. 4 — Upstream transportation · Cat. 9 — Downstream transportation

End-of-Life (Beyond Today)

Cat. 11 — Use of sold products · Cat. 12 — End-of-life treatment

When Supplier Engagement Works at Scale: Project Gigaton



Walmart's Project Gigaton (Launched 2017)

Walmart invited its more than 100,000 suppliers to voluntarily reduce or avoid 1 billion metric tons (a gigaton) of greenhouse gas emissions from its supply chain by 2030 — targeting energy, waste, packaging, transportation, nature, and product design.

The Result

In February 2024, Walmart announced suppliers had collectively reported over 1 billion metric tons reduced or avoided — six years ahead of the 2030 target.

The Scale

More than 5,000 suppliers ultimately participated — described as one of the largest private-sector supply chain sustainability consortia ever assembled.

The Method

Not top-down mandates — a shared toolkit, education, financing support, and public recognition for supplier progress.

The Honest Caveat

The same year, Walmart disclosed its own direct (Scope 1 & 2) operational emissions rose 3.9% — supply chain success and operational progress are not the same thing.

Technology for Supply Chain Traceability

The engineering opportunity underneath everything in this lecture — visibility itself is a data problem.

Blockchain-Based Traceability

Distributed ledgers can record a product's custody chain immutably, from raw material to shelf, making origin claims independently verifiable.

Satellite & Remote Sensing

Monitoring platforms can detect deforestation or land-use change linked to specific supply chain regions in near real time.

Supplier Data Platforms

Software that aggregates supplier-reported emissions, labor, and safety data into a single auditable system, replacing scattered spreadsheets.

Why This Is Your Opportunity

Every technology on this slide is still immature relative to the scale of the problem — this is one of the most under-built areas in all of climate tech.

Why does supply chain visibility typically collapse beyond Tier 1 suppliers?

- A Tier 2 and Tier 3 suppliers are always located in different countries
- B Companies have direct contracts only with Tier 1 suppliers, and visibility depends on those suppliers disclosing their own suppliers in turn ✓**
- C Tier 2 and Tier 3 suppliers are legally required to remain anonymous
- D Visibility does not actually decrease at any tier

ANSWER & WHY

B. Companies have direct contracts only with Tier 1 suppliers, and visibility depends on those suppliers disclosing their own suppliers in turn

A company's direct contractual relationship is with its Tier 1 suppliers. Seeing further upstream requires those suppliers to voluntarily disclose their own suppliers — a chain of disclosure that gets weaker, and less enforceable, at every additional tier.

One Chain, Three Interventions, One Measurement System

Sourcing, production, and distribution are not separate sustainability projects — they are three points on the same value chain, measured by the same framework.

Sustainable Sourcing

Reduces risk and impact before a material ever enters the factory — certifications and audits are the tools.

Sustainable Production

Reduces impact per unit made — Life Cycle Assessment is the tool that makes design choices comparable.

Sustainable Distribution

Reduces impact per unit moved — mode shift is the single highest-leverage lever available.

Measurement & Visibility

Ties all three together into one Scope 3 inventory — and is the actual bottleneck most companies face today.

"Pick a product you own. Trace it backward: where was it sourced, produced, and distributed from — and which of today's three interventions would improve it most?"

This is exactly how a real supply chain sustainability audit begins.

📄 4 MIN

Sustainable Supply Chains, Recapped

1

Sustainable Sourcing

Where materials come from — addressed through certifications like Fairtrade, FSC, and Rainforest Alliance.

2

Rana Plaza

A 2013 sourcing oversight failure that killed 1,134 people and reshaped global supplier auditing.

3

Sustainable Production

Cleaner production prevents waste at the source — measured through Life Cycle Assessment.

4

LCA Boundaries

Cradle-to-gate, cradle-to-grave, and cradle-to-cradle — each a progressively wider measurement scope.

5

Material Substitution

A single virgin-to-recycled material swap saved 115 tonnes CO₂e/year in our worked example — at unchanged product design.

6

Sustainable Distribution

Often the industry's rare true win-win: lower emissions and lower cost, via the same lever.

7

Modal Shift

Sea freight can be roughly 60× less emissions-intensive than air freight for an identical shipment.

8

Supply Chain Tiers

Visibility collapses sharply beyond Tier 1 — the central operational challenge of this entire field.

9

Project Gigaton

Walmart's supplier engagement model hit a 1-gigaton target six years early — while its own operational emissions still rose.

10

The Scope 3 Connection

Sourcing, production, and distribution map directly onto Scope 3 Categories 1–2, 3 & 5, and 4 & 9.

OPERATIONAL SUSTAINABILITY

Resource Management

Energy Efficiency · Water Conservation · Waste Reduction · Circular Economy Principles

**The global economy is only 6.9% circular.
The other 93% is resources used once, then lost.**

*Today is entirely practical: the actual calculations engineers and facility managers use to cut energy, water, and waste
— and the systems-level thinking that ties them together.*

Four Resources, One Systems Problem



01

Energy Efficiency

Doing the same work with less power — and calculating whether it pays for itself.



02

Water Conservation

Protecting the resource India is running out of fastest.



03

Waste Reduction

The hierarchy of what to do with material before it becomes garbage.



04

The Circular Economy

The systems-level model that ties all three together permanently.

SECTION 1 OF 4

Energy Efficiency

Delivering the same output, service, or comfort — while using measurably less power.



What Is Energy Efficiency?

A precise engineering concept, often confused with its cousin, energy conservation.

The Working Definition

Using less energy to provide the same product, service, or level of comfort — doing the same job, with less input.

Efficiency vs. Conservation

Conservation reduces energy use by reducing the service itself (turning off a light); efficiency reduces energy use while keeping the service the same (a more efficient light, on for the same hours).

Energy Intensity: The Macro View

Energy used per unit of economic output (e.g., MJ per unit of GDP) — a country-level indicator of how efficiently an economy converts energy into value.

Four Levers of Energy Efficiency

The same underlying logic — do more with less — applied across every major energy-consuming system.



Buildings & HVAC

Insulation, smart thermostats, and efficient heating/cooling — buildings account for a large share of global energy demand.



Industrial Processes

Waste heat recovery, efficient motors, and process optimization — often the fastest payback in an industrial facility.



Lighting

LED lighting uses a small fraction of the energy of incandescent bulbs for the same brightness — one of the simplest, fastest wins available.



Transport & Logistics

Fuel efficiency standards, route optimization, and electrification reduce energy use per kilometre travelled.

ISO 50001: Energy Management Systems

Efficiency gains stick when they're built into a management system — not left to one-off projects.

What It Is

The international standard for establishing, implementing, and continually improving an energy management system within an organization.

The Core Cycle

A Plan-Do-Check-Act cycle: set an energy baseline and targets, implement improvements, measure results, and adjust — repeated continuously.

Why It Matters

It turns "we should save energy" into an auditable, repeatable organizational process — the same logic industrial quality systems use for defects.

Payback Period on an LED Lighting Retrofit

GIVEN

- 200 incandescent bulbs (60 W each), operating 10 hours/day
- Replacing with 200 LED bulbs (9 W each)
- Electricity cost: ₹7/kWh; Retrofit cost: ₹250/bulb (₹50,000 total)

FORMULA

$$\text{Payback Period (years)} = \text{Total Investment} \div \text{Annual Cost Savings}$$

STEP-BY-STEP CALCULATION

- 1 Power saved per bulb: $60\text{W} - 9\text{W} = 51\text{W} = 0.051\text{ kW}$
- 2 Daily savings (200 bulbs): $0.051\text{ kW} \times 10\text{ hrs} \times 200 = 102\text{ kWh/day} \rightarrow 37,230\text{ kWh/year}$
- 3 Annual cost savings: $37,230\text{ kWh} \times ₹7/\text{kWh} = ₹2,60,610/\text{year}$
- 4 Payback: $₹50,000 \div ₹2,60,610 \approx 0.19\text{ years} \approx 2.3\text{ months}$

≈ 2.3 Months

The retrofit pays for itself in under three months — incandescent-to-LED conversions are among the fastest-payback efficiency investments that exist.

"Of the four levers — buildings, industry, lighting, transport — which do you think offers the fastest payback for a typical Indian college campus, and why?"

Think about what's cheapest to retrofit versus what saves the most energy.

🕒 3 MIN

What is the key difference between energy efficiency and energy conservation?

- A They are exactly the same thing
- B Efficiency keeps the service the same while using less energy; conservation reduces the service itself ✓**
- C Conservation only applies to industry, efficiency only to buildings
- D Efficiency always costs more than conservation

ANSWER & WHY

B. Efficiency keeps the service the same while using less energy; conservation reduces the service itself

Energy conservation reduces energy use by reducing or eliminating the service (e.g., turning off a light). Energy efficiency reduces energy use while keeping the exact same service level (e.g., a more efficient light, on for the same number of hours) — a subtle but important engineering distinction.

SECTION 2 OF 4

Water Conservation

Protecting the resource that India, specifically, is running out of fastest.



India's Water Stress, By The Numbers

~600 M

Indians facing high to extreme water stress (NITI Aayog, CWMI)

820 M

People live in river basins with per capita availability at or below the scarcity threshold

1,486 m³

India's per capita water availability (2021), projected to fall toward 1,140 m³ by 2050

18% / 4%

Share of world's population India holds, versus share of world's freshwater resources

India's water demand is projected to be twice its available supply by 2030 — an estimated 6% loss of GDP if unaddressed.

Source: NITI Aayog, Composite Water Management Index (CWMI); Ministry of Jal Shakti.

Four Tools for Water Conservation

Interventions available at every scale, from a single building to an entire watershed.



Low-Flow Fixtures

Efficient taps, showerheads, and dual-flush toilets can cut household and institutional water use significantly with no change in service.



Rainwater Harvesting

Capturing rooftop and surface runoff for direct use or groundwater recharge — cheap, decentralized, and highly effective at scale.



Wastewater Recycling

Treating and reusing greywater and treated sewage for irrigation, cooling, or flushing — reducing demand on freshwater sources.



Precision Irrigation

Drip and micro-irrigation target water directly at crop roots — critical since agriculture consumes roughly 85% of India's water use.

Calculating Rainwater Harvesting Potential

GIVEN

- Rooftop catchment area: 500 m²
- Average annual rainfall in the region: 900 mm (0.9 m)
- Runoff coefficient for a concrete roof: 0.85 (85% collection efficiency)

FORMULA

$$\text{Harvestable Water (litres)} = \text{Area (m}^2\text{)} \times \text{Rainfall (m)} \times \text{Runoff Coefficient} \times 1000$$

STEP-BY-STEP CALCULATION

- 1 Volume of rain falling on the roof: $500 \text{ m}^2 \times 0.9 \text{ m} = 450 \text{ m}^3$
- 2 Apply the runoff coefficient: $450 \text{ m}^3 \times 0.85 = 382.5 \text{ m}^3$
- 3 Convert to litres: $382.5 \times 1,000 = 3,82,500 \text{ litres}$

3.83 Lakh L/Year

This single rooftop can realistically harvest nearly 4 lakh litres of water annually — enough to meaningfully offset a building's freshwater demand.

Jal Shakti Abhiyan

India's flagship national water conservation campaign, launched in 2019.

The Mandate

A time-bound, mission-mode campaign focused on water-stressed districts, run jointly by central and state governments.

The Five Focus Areas

Water conservation and rainwater harvesting; renovation of traditional water bodies; reuse and recharge structures; watershed development; and intensive afforestation.

Why It Matters Here

It operationalizes the exact interventions in this section — rainwater harvesting, recharge, and precision water management — at national scale.

Roughly what share of India's total water use goes to agriculture, making precision irrigation such a high-leverage intervention?

A About 25%

B About 50%

C About 85%



D About 10%

ANSWER & WHY

C. About 85%

Agriculture accounts for roughly 85% of India's total water consumption, primarily through irrigation — making even modest efficiency gains from drip and precision irrigation enormously significant at national scale.

SECTION 3 OF 4

Waste Reduction

The hierarchy of what to do with material — long before it becomes something to throw away.



The Waste Hierarchy

Not all waste management options are equal — this hierarchy ranks them from most to least preferred.

1. Refuse & Reduce

Avoid creating the waste in the first place — the only tier that eliminates impact entirely.

2. Reuse

Use a product again, in its original form, for the same or a different purpose.

3. Recycle

Break a material down and remanufacture it into something new.

4. Recover

Extract energy from waste that cannot be reused or recycled (e.g., waste-to-energy).

5. Dispose

Landfill — the least preferred option, used only when nothing else is possible.

SECTION 3 • WORKED EXAMPLE

Calculating a City's Landfill Diversion Rate

GIVEN

- A city generates 1,000 tonnes of municipal solid waste per day
- Of this: 350 tonnes is composted, 150 tonnes is recycled
- The remaining 500 tonnes goes to landfill

FORMULA

$$\text{Diversion Rate (\%)} = (\text{Waste Diverted from Landfill} \div \text{Total Waste Generated}) \times 100$$

STEP-BY-STEP CALCULATION

- 1 Total waste diverted: composted + recycled = 350 + 150 = 500 tonnes
- 2 Diversion rate: $(500 \div 1,000) \times 100 = 50\%$

50% Diversion

Exactly half of this city's waste is being kept out of the landfill — the standard metric cities and campuses use to track waste management progress.

Extended Producer Responsibility (EPR)

A policy principle that shifts the cost and responsibility for waste back onto the companies that created it.

The Core Idea

Producers are made financially and physically responsible for the end-of-life management of the products and packaging they place on the market.

India's E-Waste (Management) Rules, 2022

Mandate producers of electronics to meet specific e-waste collection and recycling targets, obtained through registered recyclers.

India's Plastic Waste Management Rules

Require producers, importers, and brand owners of plastic packaging to meet EPR targets for collection and recycling.

Why It Changes Behaviour

When producers bear disposal costs, they have a direct financial incentive to design products that are easier and cheaper to recycle.

Where Does This Sit in the Waste Hierarchy?

For each real action, decide which tier of the waste hierarchy it belongs to.

		REDUCE/REUSE	RECYCLE/RECOVER	DISPOSE
1	Refilling a glass bottle instead of buying a new one	1	2	3
2	Melting down aluminium cans to make new cans	4	5	
3	Sending unsorted mixed waste directly to a landfill	6		
4	Designing packaging to use less material from the start			
5	Burning non-recyclable waste to generate electricity			
6	Donating used furniture instead of throwing it away			

Under Extended Producer Responsibility (EPR), who is made financially responsible for a product's end-of-life management?

- A The consumer who purchased the product
- B The local municipal government
- C The producer or brand owner who placed the product on the market** ✓
- D Recycling companies exclusively

ANSWER & WHY

C. The producer or brand owner who placed the product on the market

EPR shifts responsibility for end-of-life management — collection, recycling, safe disposal — back onto the producer or brand owner, creating a direct financial incentive to design products that are easier and cheaper to recycle.

SECTION 4 OF 4

The Circular Economy

The systems-level model that ties energy, water, and waste together — permanently, not as separate projects.



Three Principles of a Circular Economy

As defined by the Ellen MacArthur Foundation — the most widely cited framework in the field.



Design Out Waste & Pollution

Waste is treated as a design flaw, not an inevitability — products and systems are designed from the outset to prevent it.



Keep Products & Materials in Use

Through reuse, repair, remanufacturing, and recycling — extending value for as long as physically possible.



Regenerate Natural Systems

Circular systems actively rebuild natural capital — returning nutrients to soil, restoring ecosystems — not just minimizing harm.

The Technical Cycle vs. the Biological Cycle

The Ellen MacArthur Foundation's "butterfly diagram" splits circularity into two distinct loops, based on what kind of material is involved.



The Technical Cycle

- Applies to durable, manufactured materials: metals, plastics, glass, electronics.
- Circulates through maintenance, reuse, refurbishment, remanufacturing, and recycling.
- Goal: keep the material in a manufactured loop indefinitely, never returning to raw extraction.



The Biological Cycle

- Applies to renewable, biodegradable materials: food waste, natural fibers, wood.
- Circulates through composting and anaerobic digestion, safely returning nutrients to the soil.
- Goal: regenerate natural systems, not just avoid landfill.

The Circularity Gap

6.9%

The share of materials in the global economy that are cycled back — down from 9.1% in 2018

93.1%

Share of all materials entering the global economy that still come from virgin extraction

€25.4 Tn

The "Value Gap": economic value lost annually to linear resource use — nearly 31% of global GDP

Despite two decades of circular economy advocacy, the world has gotten measurably less circular, not more — the gap is the opportunity.

Source: Circle Economy & Deloitte, Circularity Gap Report 2025 & 2026.

The Five Pathways of Value Loss

The Circularity Gap Report 2026 breaks the €25.4 trillion Value Gap into five specific leaks in the system.



Technical Cycle or Biological Cycle?

For each material, decide which of the two circular economy loops it belongs to.

- | | |
|---|---------------------------------|
| 1 | Aluminium cans |
| 2 | Food scraps and vegetable peels |
| 3 | Smartphone components |
| 4 | Garden and yard waste |
| 5 | Glass bottles |
| 6 | Untreated wood offcuts |

TECHNICAL CYCLE

BIOLOGICAL CYCLE

1

2

3

4

5

6

According to the Circularity Gap Report, what is the single largest of the five pathways of value loss in the global economy?

A Food losses and waste

B Processing losses

C End-of-life waste



D Premature deterioration of buildings

ANSWER & WHY

C. End-of-life waste

End-of-life waste — products landfilled or incinerated instead of refurbished, reused, or recycled — is the largest single pathway of value loss, estimated at roughly €10.0 trillion annually in the Circularity Gap Report 2026.

One Systems Problem, Not Four Separate Ones

Energy, water, and waste are not independent projects — the circular economy is the framework that shows why.

Energy Efficiency

Reduces the energy embedded in every material extracted, processed, and moved — the "energy losses" pathway in the Value Gap.

Water Conservation

Recycled water and precision irrigation are themselves circular loops — keeping a resource in use instead of extracting more.

Waste Reduction

The waste hierarchy is a miniature version of the circular economy — refuse and reuse before recycle, recycle before dispose.

The Circular Economy

The framework that shows these three are not separate initiatives — they are one system, viewed from three different points.

"Pick a building or campus you know well. Where would you find the fastest-payback energy fix, the biggest water-saving opportunity, and the worst waste leak — all three?"

This is exactly how a resource management audit actually starts.

📄 4 MIN

Resource Management, Recapped

1

Energy Efficiency

Same service, less energy — distinct from conservation, and formalized through ISO 50001.

2

LED Payback

A retrofit that pays for itself in months, not years — the fastest lever in this lecture.

3

India's Water Stress

~600 million facing high-extreme stress; demand set to outpace supply 2× by 2030.

4

Rainwater Harvesting

A single rooftop can realistically harvest ~3.8 lakh litres a year — decentralized and cheap.

5

The Waste Hierarchy

Refuse and reduce first; disposal to landfill is always the last resort, not the default.

6

Landfill Diversion Rate

The standard metric — composted plus recycled, divided by total waste generated.

7

Extended Producer Responsibility

Shifts end-of-life cost back onto producers, incentivizing recyclable design from the start.

8

Three Circular Economy Principles

Design out waste, keep materials in use, regenerate natural systems (Ellen MacArthur Foundation).

9

The Circularity Gap

The world is only 6.9% circular — and getting less circular, not more, despite growing awareness.

10

One System

Energy, water, and waste are the same underlying resource-management problem, viewed from three angles.